Gabriel Gallo

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Education

University of California, Berkeley Dec 2026
Bachelor of Science in Electrical Engineering and Computer Science *GPA: 3.66/4.00*

- **Coursework:** Discrete Mathematics and Probability, Introduction to Full-Stack Development, Data Structures and Algorithms, Introduction to Robotics, Digital Design and Integrated Circuits, Computer Security

Technical Skills

Languages: Python, SQL, C/C++, Java, JavaScript, TypeScript, HTML/CSS, SystemVerilog, Assembly, Go
Framework & Libraries: React.js, Node.js, Express.js, NumPy, MongoDB, Django, ROS2
Developer Tools: Git, GitHub Copilot, Cursor, Cloudflare, Firebase, AWS, Netlify

Experience

Software Engineer May 2025 – Aug. 2025
UNIFIBD *San Jose, CA*

- Led the end-to-end development of a cross-platform mobile application supporting individuals with Crohn's disease, reaching over 5000 users.
- Migrated legacy Swift iOS app to Flutter, consolidating iOS/Android into a single codebase and doubling platform reach.
- Published and maintained the app on the Apple App Store and Google Play Store, increasing accessibility and visibility across platforms.
- Optimized app performance by reducing load times by 30%, leading to a smoother and more responsive user experience.
- Collaborated with a team of 3 engineers, implementing platform-specific enhancements and conducting cross-device testing to ensure a consistent UX across Android and iOS.

Projects

RISC-V 3-Stage Pipelined CPU | *SystemVerilog, Xilinx Vivado, PYNQ (AUP-ZU3)*

- Designed and implemented a 3-stage pipelined RISC-V processor in SystemVerilog, targeting a Zynq UltraScale+ FPGA.
- Resolved data and control hazards in the pipeline to achieve a target CPI < 1.2 and clock frequency > 100 MHz.
- Integrating a UART interface for BIOS tethering and hardware-in-the-loop debugging using Xilinx ILA.
- Implemented support for the Zephyr RTOS shell, demonstrating system-level integration of hardware and software.

Flow | *Angular, SCSS, TypeScript, Python, Django, MongoDB, AWS, Gemini API*

- Switch seamlessly between email and calendar in an authenticated dashboard with collapsible sidebar navigation
- Secure Auth: Integrated Google OAuth 2.0 login via django-allauth.
- AI-Powered Email Tools: Backend processes email content for summarization, categorization, and event detection using NLP (spaCy, NLTK)
- Built with Django, Django REST Framework, and PostgreSQL, using Celery and Redis for background task processing

TurtleBot Interceptor | *Python, ROS2, OpenCV, YOLO, SLAM, NumPy*

- Developed Python/OpenCV cone detector with color/shape filtering and distance estimation; improved detection precision from 78% to 93% in varied indoor lighting.
- Engineered MPC controller integrating Extended Kalman Filter, Monte Carlo Localization, local grids, and cone detections; reduced intercept time by ~1.2 s and cut path oscillations by ~30%.
- Instrumented visualization for cones, MPC trajectories, and obstacles; cut debugging time during field tests by ~25%.

Secure File Storage System | *Golang, Cryptography (AES-CTR, RSA-OAEP, HMAC-SHA512), Integration Testing*

- Architected a decentralized, stateless file storage client in Go utilizing an untrusted remote datastore, maintaining total confidentiality and integrity against a global network adversary.
- Designed a dynamic multi-user sharing hierarchy utilizing a centralized Sharing Tree map, successfully decoupling file appends from metadata growth to satisfy a strict bandwidth-efficiency requirement.
- Authored comprehensive integration and tampering test suites validating complex edge cases, including empty files, multi-device synchronization, and structural datastore corruptions.